PART 02 · CHAPTER 06

Support, resistance & price areas

Map prior reactions as conditional evidence, with explicit boundaries.

Answer edition: includes continuation, worked solutions and the knowledge-check answer.

beginner About 5 min of prose · practice takes additional time

Draft manuscript · source notes below

> THE IDEA TO KEEP
>
> A level is a predefined area to investigate. It is neither an invisible wall nor a guaranteed execution price.

By the end of this chapter

- Construct a shaded price zone from named prior reactions.
- Freeze the zone before examining later observations.
- Apply a close-based buffer without silently changing it.
- Distinguish entering a zone, reacting, and crossing a failure condition.

Builds on: Trends, swings & changing structure

First pass: freeze the D12 map

Work only from the construction data below before opening the chapter continuation. This is an attempt followed by a reference explanation; write your own map and conditions first.

Use the one-before/one-after closing-price trough rule. Select recognized troughs through D12, cluster them under a maximum $0.60 span, and freeze the zone edges. Define a visit, a $1 rise within two sessions after a visit, and a strictly-below break using a $0.30

downward buffer. State when each could be known and what would challenge the map. Do not invent D13 onward.

Construction closes, available through D12

Session	Close ($)
D1	43
D2	41
D3	39.8
D4	42
D5	45
D6	41
D7	40.1
D8	43
D9	45
D10	42
D11	40.35
D12	43

Compare the continuation with your frozen note. This reference includes outcome headings, complete ledgers and solutions. In the learner print edition it starts on a new page after the cutoff packet.

Start with reactions that can be located

Support names an area associated with prior interrupted declines or a conditional expectation of another such reaction. Resistance names the analogous area for interrupted advances. The first part is observable history; the expectation is a hypothesis. We will build a support map from the same supplied closes that appear in the construction panel, then reveal later closes separately. This sequence makes the analytical choice visible.

Use a simple one-before/one-after closing-price trough rule for this mapping exercise. It differs from chapter 5's two-bar rule because the present question is a small reaction cluster, and the supplied table explicitly declares that change. We do not change it mid-example. A trough must be strictly below its immediate neighbors and becomes recognized after the next close. This produces three selected construction troughs at D3$39.80, D7$40.10, and D11$40.35.

Construction closes, available through D12

Session	Close ($)
D1	43
D2	41
D3	39.8
D4	42
D5	45
D6	41
D7	40.1
D8	43
D9	45
D10	42
D11	40.35
D12	43

Select the observations and explain the exclusions

D3 is below D2$41 and D4$42, D7 is below D6$41 and D8$43, and D11 is below D10$42 and D12$43. All three follow declines and are followed by higher closes. At D12 every selected trough is recognized. The intermediate $41 close at D6 is excluded because the next $40.10 is lower; D10$42 is excluded for the same reason. These exclusions prevent every near-$40 mark from becoming a separate “touch.”

For this exercise, cluster the recognized troughs when their high-minus-low span is no more than $0.60. The observed span is 40.35 −39.80 = $0.55, so the three qualify. Choose the zone’s edges as the minimum and maximum selected troughs, $39.80–$40.35. The width is $0.55, or 1.375% of an illustrative $40 reference. The cluster tolerance is an original teaching choice, while the resulting edges are transparent arithmetic.

A reasonable alternative is to draw one reference at the cluster’s median $40.10. Another analyst could use intraday lows instead of closes and get different boundaries. Those are separate constructions. Our closing-price zone is chosen because its inputs and recognition times are supplied; it does not claim to reveal all resting orders.

Construction decisions frozen after D12

Selected trough	Previous close ($)	Trough close ($)	Next close ($)	Known after
D3	41	39.8	42	D4
D7	41	40.1	43	D8
D11	42	40.35	43	D12
Zone	Minimum trough	39.8	Maximum trough 40.35	D12

Write the reaction and failure definitions before revealing

For our closing-price exercise, a visit is a close inside the frozen zone. A reaction is a visit followed within the next two completed sessions by a close at least $1 above the visiting close. A buffered downward break is a completed close below $39.50, thirty cents below the zone’s lower edge. These definitions do not follow from a law of support; they make the next six observations classifiable.

Why include a buffer? It lets a slight edge crossing and a larger excursion be different events. Why $0.30? To make this example easy to inspect, not because thirty cents is universally

informative. The definition must stay fixed when a later close is uncomfortable. A stricter buffer or a two-close requirement could be examined as an alternative, but each would produce a different event set.

At D12, write this note: “Three one-bar closing troughs fit a $0.60 cluster; zone 39.80–40.35 is frozen. Watch for the defined visit/reaction, and separately record any completed close below 39.50.” You have not yet seen D13 onward. The future panel makes that separation visible instead of letting later bounces dictate where the zone should be.

Reveal a reaction, an edge crossing, and a buffered break

D13 closes at $40.30, inside the zone. D14 at $41.50 is $1.20 above it within the allowed two-session window, satisfying the reaction definition. D15 at $39.70 falls ten cents below the lower zone edge but remains above the $39.50 break threshold. D16 at $39.40 meets the buffered break. These events differ; labeling them all either “held” or “failed” would discard useful information.

D17 at $40.15 returns inside the same frozen zone after the break; D18 at $39.20 falls below the threshold again. That sequence does not erase D16’s earlier observation. A new role-reversal hypothesis could ask whether the old area now interrupts advances, but one return inside the area is not yet a measured resistant reaction. State the new condition before classifying it.

The zone remains 39.80–40.35 in both panels and in every row. If we lower its edge to 39.20 after D18, the apparent history of successful support changes because the definition changed, not because the original calculation improved. Preserve the old note and label any revised map as a new analysis.

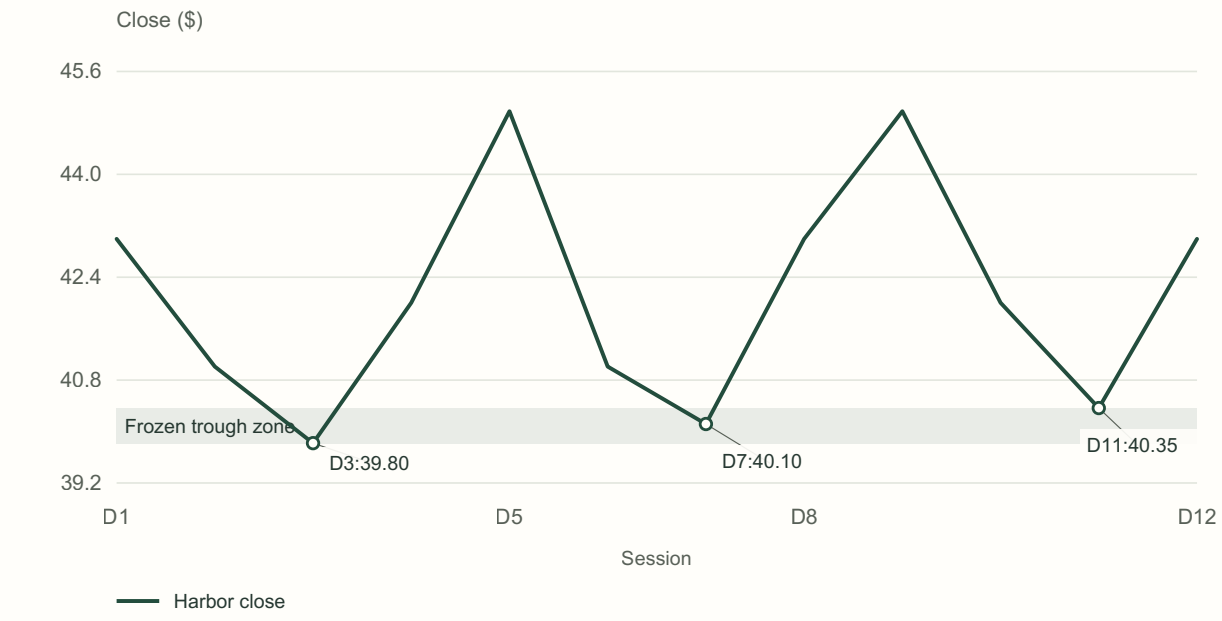
Later closes under the frozen definitions

Session	Close ($)	Classification available after that close
D13	40.3	Visit inside zone
D14	41.5	Reaction:+1.20 within two sessions
D15	39.7	Below zone edge, above buffered threshold
D16	39.4	Buffered downward break
D17	40.15	Return inside old zone
D18	39.2	Below buffered threshold again

FICTIONAL TEACHING ILLUSTRATION

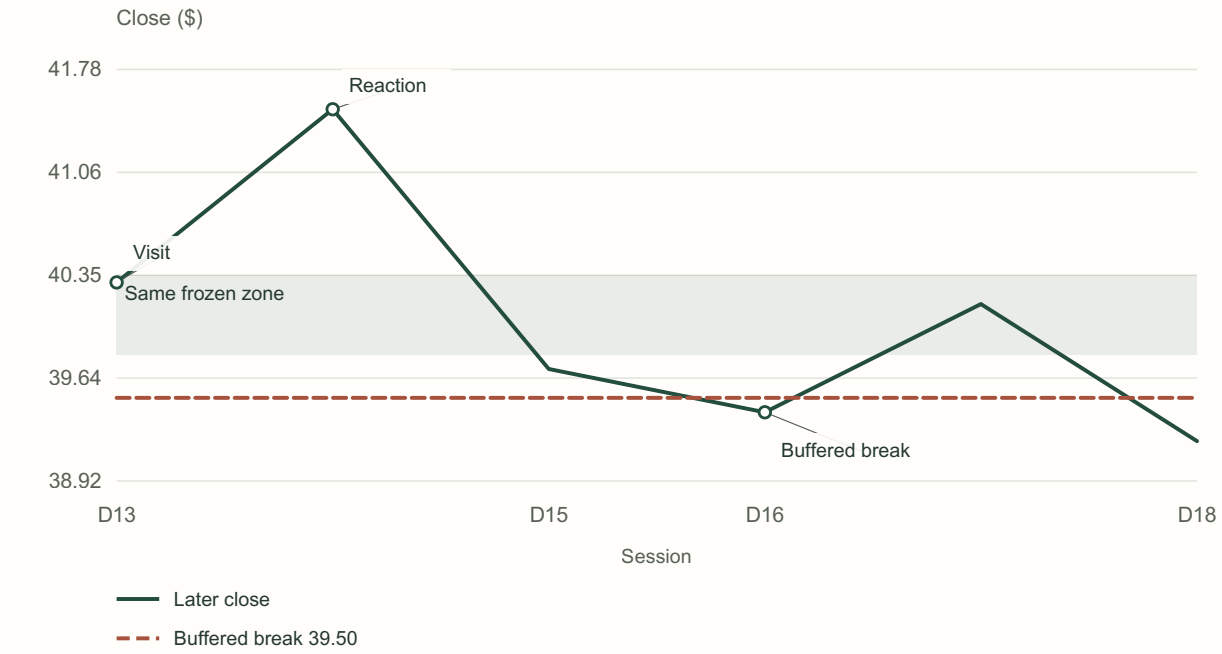
Construct the zone, then reveal new observations

Construction: only D1–D12



Fictional supplied observations.

Reveal: D13–D18



Fictional supplied observations.

One closing-price cluster, frozen after D12. Both panels use the same 39.80–40.35 shaded zone. The lower 39.50 reference is a separately declared buffered-break condition, not a moved zone edge.

View the illustration’s data

Construction: only D1–D12

Session	Harbor close
D1	43
D2	41
D3	39.8
D4	42
D5	45
D6	41
D7	40.1
D8	43
D9	45
D10	42
D11	40.35
D12	43

Reveal: D13–D18

Session	Later close	Buffered break 39.50
D13	40.3	39.5
D14	41.5	39.5
D15	39.7	39.5
D16	39.4	39.5
D17	40.15	39.5
D18	39.2	39.5

A low-based rule can disagree with a close-based rule

The extra candle for D15 has O40.20/H40.40/L39.30/C39.70. Its intraday low is below $39.50, while its completed close is above $39.50. An intraday-trigger instruction would detect

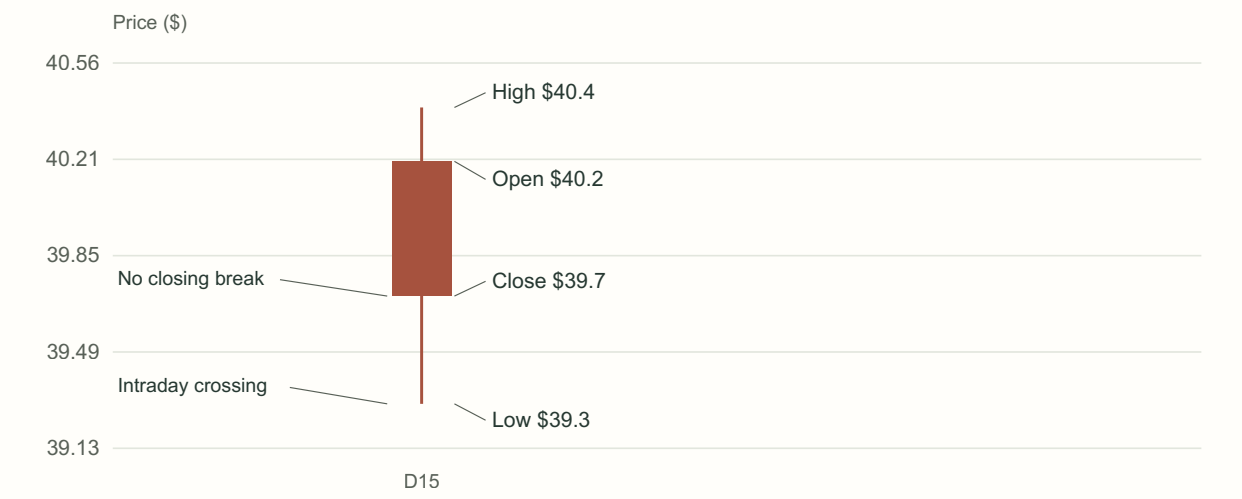
the excursion; our analytical close-based break rule would not. Both can be applied correctly to the same candle because they ask different questions.

An analytical invalidation rule tells you when evidence challenges a hypothesis. An order instruction tells a broker how to attempt a transaction. If your rule waits for a completed close, a stop-market order at the intraday level is a different operational choice. Keep that distinction in the plan, including what you do during a gap. A line or zone is not an execution promise.

For a $41 assumed entry and a $39.50 trigger, price distance is $1.50 per share. If a later feasible exit is $37, price loss is $4 per share instead. Chapter 1 taught why sizing and larger-loss scenarios accompany the hypothesis. The integrated breakout case will use a complete set of these choices rather than treating a zone's width as a maximum possible loss.

FICTIONAL TEACHING ILLUSTRATION

D15 crosses the intraday threshold, but not the close rule



Body: $0.50. Upper shadow: $0.20. Lower shadow: $0.40.

D15 completed candle

D15's 39.30 low lies below 39.50; its 39.70 close remains above it. The closing-price zone and buffer retain their original definitions.

View the illustration's data

D15 crosses the intraday threshold, but not the close rule

Interval	Open	High	Low	Close
D15	40.2	40.4	39.3	39.7

A small number of areas beats an all-purpose drawing

The construction above gives a reason for including one zone and excluding several nearby marks. A resistance exercise uses interrupted advances instead: select declared local peaks, cluster their prices under a stated span tolerance, and freeze the resulting area before testing the next visit. The independent practice gives you a fresh example. This symmetry helps you build a map rather than memorize one picture.

If every price gets a line, almost every turn can be explained afterward as near something. A useful map is sparse enough to be challenged. Count a visit only under the declared field and

cutoff, then describe whether the reaction definition occurred. The more careful wording is “these prior closing troughs form this area” and “the later data met or did not meet this condition.” Supply-and-demand explanations can remain hypotheses while the measurements are precise.

A plausible mechanism has a market boundary

Osler’s order data provide a concrete example of round-number clustering in currency markets. They do not reveal the orders near Harbor’s fictional zone or test a stock support rule. Keep the mechanism as a hypothesis until the relevant stock data and executable rule have been evaluated.

A WORKED EXAMPLE

A completed map and an honest update

The D3/D7/D11 closing troughs create 39.80–40.35 after D12. D13–D14 satisfy the declared visit/reaction. D15 crosses the zone edge without meeting the completed-close 39.50 break; D16 meets it. Later closes are judged against the same frozen definition. This provides a successful reaction and a subsequent challenge without silently changing the drawing.

Pause & try it

Map the three troughs at 39.80, 40.10, and 40.35. Define a break with an explicit field, buffer, and observation time. Write one scenario in which an intraday crossing and a close-based rule disagree.

Draw a shaded band with lower edge 39.80 and upper edge 40.35; mark the three supplied troughs inside it. Its span is 0.55. Under the chapter’s 0.60 cluster tolerance, all qualify.

One complete break rule is a completed regular-session close below 39.50, a 0.30 buffer beneath 39.80. D16’s 39.40 meets it. The strict word “below” means a close exactly 39.50 does not.

D15 O40.20/H40.40/L39.30/C39.70 crosses 39.50 intraday but does not finish below it. Alternative fields, buffers or repeated-close rules are defensible if written before the later data and applied consistently.

GUIDED PRACTICE

Guided: stop before the reveal

Complete this partly filled worksheet under the supplied clustering rule: selected peaks are $55.10, $54.90 and $55.30; span = $0.40; zone = [____, ____]. Use a $0.20 outward buffer. Classify later closes $55.20, $55.40 and $55.60, showing the break threshold before revealing them.

The selected peaks give zone [54.90, 55.30]. The upward buffer places the threshold at 55.30 + 0.20 = $55.50; a buffered break requires a completed close strictly above it.

A $55.20 close visits the zone. $55.40 crosses its upper edge but does not meet the buffered-break rule. $55.60 is a buffered upward break. No later reversal or executable fill follows from these classifications alone.

INDEPENDENT PRACTICE

Independent: construct a resistance cluster

Recognized closing-price peaks are $72.40, $72.10 and $72.60. Use a maximum cluster span $0.60. Freeze edges and a $0.25 upward buffer. Classify later completed closes $72.50, $72.75, $72.85 and $72.90; state how exact equality at the threshold is treated.

Span = 72.60 − 72.10 = $0.50, within the $0.60 tolerance. Zone = 72.10–72.60; threshold = 72.60 + 0.25 = $72.85.

72.50 visits the zone; 72.75 crosses the upper edge without a buffered break. 72.85 equals the threshold and does not satisfy strictly above. 72.90 meets the buffered-break condition. Keep the field and comparison fixed before the later closes.

CHECK YOUR UNDERSTANDING

One question before you go

The frozen zone is 39.80–40.35 and break rule is a completed close below 39.50. D15 has low 39.30 and close 39.70. Which condition occurred?

A. A buffered closing break, because the low was below 39.50

B. No edge crossing, because the close was near 40

C. An edge crossing but no buffered closing break

Answer: C. 39.70 is below the 39.80 zone edge but above the 39.50 closing threshold.

Sources & further reading

- Fidelity · Support and resistance ↗
  Support/resistance vocabulary and the conventional role-reversal hypothesis; not a guarantee of future reactions. Access recorded 2026-10-01.
- SEC / Investor.gov · Types of orders ↗
  Market, limit, and stop order distinctions; execution prices and fills are not guaranteed by a chart level. Access recorded 2026-10-01.
- SEC / Investor.gov · Understanding Order Types ↗
  Market/limit/stop/stop-limit execution, nonfills, trigger conventions, IOC cancellation; definitions only, not assumed fill probabilities. Access recorded 2026-10-01.
- StockCharts ChartSchool · Support and resistance ↗
  Breaks of previously defined trading areas, approximate zones, role-change and failed-crossing vocabulary. The numerical breakout and failure rules in this book are original fictional examples. Access recorded 2026-10-01.
- Osler (2001), Currency Orders and Exchange-Rate Dynamics, Staff Report 125 ↗
  FX order clustering mechanism; one dealing bank and three currency pairs, not stock-strategy evidence. Access recorded 2026-10-01.

Teaching examples are fictional unless labelled historical. Inspect this chapter’s evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

Trends, swings & changing structure ↗ Breakouts, pullbacks & failed moves ↗

Risk & position sizing ↗

Practice & review this part →